AI RISK MANAGEMENT SYSTEM

Beginner-Friendly Infographic User Guide

Use this guide when testing, presenting, or explaining the MERN project to a person who is seeing it for the first time.

Project Flow: From Login to Risk Report

1 **Login** Seed user → 2 **Project** Create scope → 3 **Sprint** Velocity plan → 4 **Backlog** Tasks + bugs → 5 **AI Scan** Detect risks → 6 **Register** Mitigate → 7 **Reports** Show result

Goal: identify risks early, plan mitigation, monitor warning signs, and manage contingency actions.

What the project does

01 Proactive risk control Find risks before they become failures.	**02 Agile tracking** Create projects, sprints and backlog items.	**03 AI-style detection** Rules check overdue tasks, bugs, velocity and workload.
04 Risk Register Store risks with probability, impact, severity and RMMM.	**05 Reports + alerts** Dashboard, notifications and reports show project health.	**06 Manual risk entry** Add vendor, customer, technology or business risks.

Roles and login credentials

Role	Login	Main job	Use these pages
Project Manager	manager@riskpro.com / password123	Create project, sprint, backlog, run AI scan, manage risks.	Projects, Sprint Board, AI Analyzer, Risk Register, Reports
Developer	dev@riskpro.com / password123	Update assigned work, bugs and blockers.	Sprint Board, Backlog, Notifications
Stakeholder	stakeholder@riskpro.com / password123	Review health, reports and alerts.	Dashboard, Reports, Notifications

Quick start commands

Backend terminal

```
cd backend
npm install
npm run seed
npm run dev
```

Frontend terminal

```
cd frontend
npm install
npm run dev
```

Use the system in this exact order

Step	What to do	Why it matters
1. Login	Use the Project Manager seed account.	Manager has permission to create and scan project data.
2. Create Project	Add project name, status, budget and dates.	This becomes the container for all sprint/risk data.
3. Create Sprint	Add target velocity and completed points.	Velocity is used to detect delay risk.
4. Add Backlog	Add tasks, stories, bugs and blockers. Select the sprint.	AI scan reads these items to detect risks.
5. Run AI Scan	Open AI Analyzer and click Scan Current Project.	Creates risks from real project data.
6. Review Risks	Open Risk Register and inspect severity and RMMM.	Shows how the manager plans action.
7. Show Reports	Open Dashboard, Notifications and Reports.	Shows project health to stakeholders.

Test data to enter

Project data	Sprint data

`Project Name: Sana AI Risk Management System`
`Status: ACTIVE`
`Budget: 50000`
`Start Date: 2026-06-01`
`End Date: 2026-08-30`
`Description: Agile project for testing schedule, quality, resource and delay risks.`
	`Sprint Name: Sprint 1 - Core Risk Module`
	`Status: ACTIVE`
	`Target Velocity: 40`
	`Completed Points: 12`
	`Start Date: 2026-06-20`
	`End Date: 2026-07-05`
	`Objective: Complete login, dashboard, sprint board and AI risk detection.`

Backlog items for generating risks

Title	Type	Priority	Due date	Points	Hours
Login API overdue issue	TASK	HIGH	2026-06-20	5	12
Dashboard UI overdue	STORY	HIGH	2026-06-21	8	14
Login validation bug	BUG	CRITICAL	2026-06-28	3	8
Risk table filter bug	BUG	HIGH	2026-06-29	3	8
Report export bug	BUG	HIGH	2026-06-30	3	8
Client approval blocker	BLOCKER	CRITICAL	2026-06-26	2	6

For every backlog item, select the same sprint and assign Sana Team Developer. Leave Blocker Reason blank except for the BLOCKER item: Stakeholder has not approved final scope, so development cannot continue.

Understand the two actions

Action	Uses real project data?	Saves risks to Risk Register?	Best use
Scan Current Project	Yes - reads project, sprint and backlog.	Yes	Final demo and proper testing.
Predict Risk	No - uses manual numbers typed into the form.	Usually simulation only	Explain how risk indicators affect risk scoring.

Prediction input guide

Input field	Meaning	Risk effect
Overdue Tasks	Tasks that crossed due date.	Schedule Risk
Bug Count	Open bug items.	Quality Risk
Overloaded Members	Developers above safe capacity.	Resource Risk
Velocity Progress	Sprint progress percentage.	Delay Risk if below target.
Task Completion	Overall completion percentage.	Delay/Schedule risk when low.
Blocker Count	Items stopping work.	Schedule/Blocker Risk
WIP Count	Work in progress count.	Resource/Delay pressure
Open Tasks	Pending/in-progress work.	Schedule pressure
Story Points Done	Completed story points.	Velocity calculation
Story Points Committed	Planned sprint points.	Velocity calculation

Suggested manual simulation values

```
Overdue Tasks: 3
Bug Count: 5
Overloaded Members: 1
Velocity Progress: 55
Task Completion: 48
Blocker Count: 2
WIP Count: 8
Open Tasks: 15
Story Points Done: 24
Story Points Committed: 58
```

How the rule-based AI engine thinks

A Schedule Risk Overdue, blocked or due-soon high-priority work.	**B Quality Risk** Open bug count reaches risk threshold.	**C Resource Risk** Assigned hours exceed safe capacity.
D Delay Risk Sprint velocity is below planned target.	**E Risk Exposure** Probability x impact creates severity score.	**F RMMM Plan** Mitigation, Monitoring and Management actions.

After risks are created

Status	Meaning	Demo action
OPEN	Risk is identified but action has not started.	Explain the risk.
MITIGATING	Manager is applying mitigation and monitoring warning signs.	Click Mitigate.
RESOLVED/CLOSED	Risk is controlled, accepted or removed.	Click Close.

Manual Risk form: why it exists

AI scan detects risks from project data, but it cannot know every external issue. Use Manual Risk for customer, vendor, technology, hardware, process, budget or business risks.

```
Risk Title: Hardware delivery may be delayed
Category: SCHEDULE
Probability: 0.6
Impact: 4
Cost Impact: 20000
Description: Project testing depends on a hardware module. If hardware delivery is delayed, testing and
final delivery will also be delayed.
Mitigation: Coordinate with hardware vendor early and prepare a software simulator for testing.
Monitoring: Track vendor delivery milestones and weekly hardware status updates.
Management: Use software simulation temporarily and revise the testing schedule.
```

Expected clean demo result

After one clean AI Scan	Avoid duplicate results
Schedule Risk Quality Risk Resource Risk Delay Risk Blocker/Schedule Risk Notifications + dashboard update	Use a fresh project. Run AI Scan once for clean output. If using an old project, delete old risks first. Do not mix old backlog data with demo data.

30-second presentation script

```
This is an AI-powered project risk management system. First, the manager creates a project, sprint and
backlog items. Developers update tasks, bugs and blockers. The AI Analyzer scans the project data and
detects schedule, quality, resource and delay risks. Each risk is saved in the Risk Register with
probability, impact, severity and an RMMM plan: mitigation, monitoring and management. The dashboard,
notifications and reports help stakeholders understand project health and take action early.
```

Common mistakes and fixes

Problem	Fix
AI Scan says 0 risks	Check that backlog items are assigned to the selected sprint/project and enough bugs, blockers or overdue data exist.
Too many repeated risks	You probably scanned an old project multiple times. Use a fresh project or delete previous risks.
Overdue risk not coming	Use due dates earlier than today or near-deadline high-priority items, depending on the version.
Login not working	Run npm run seed in backend and confirm the backend server is running.
Frontend cannot load API	Start backend and check frontend API base URL / proxy configuration.
MongoDB empty array skills	In Atlas, set skills to ["React", "Node.js", "MongoDB"].